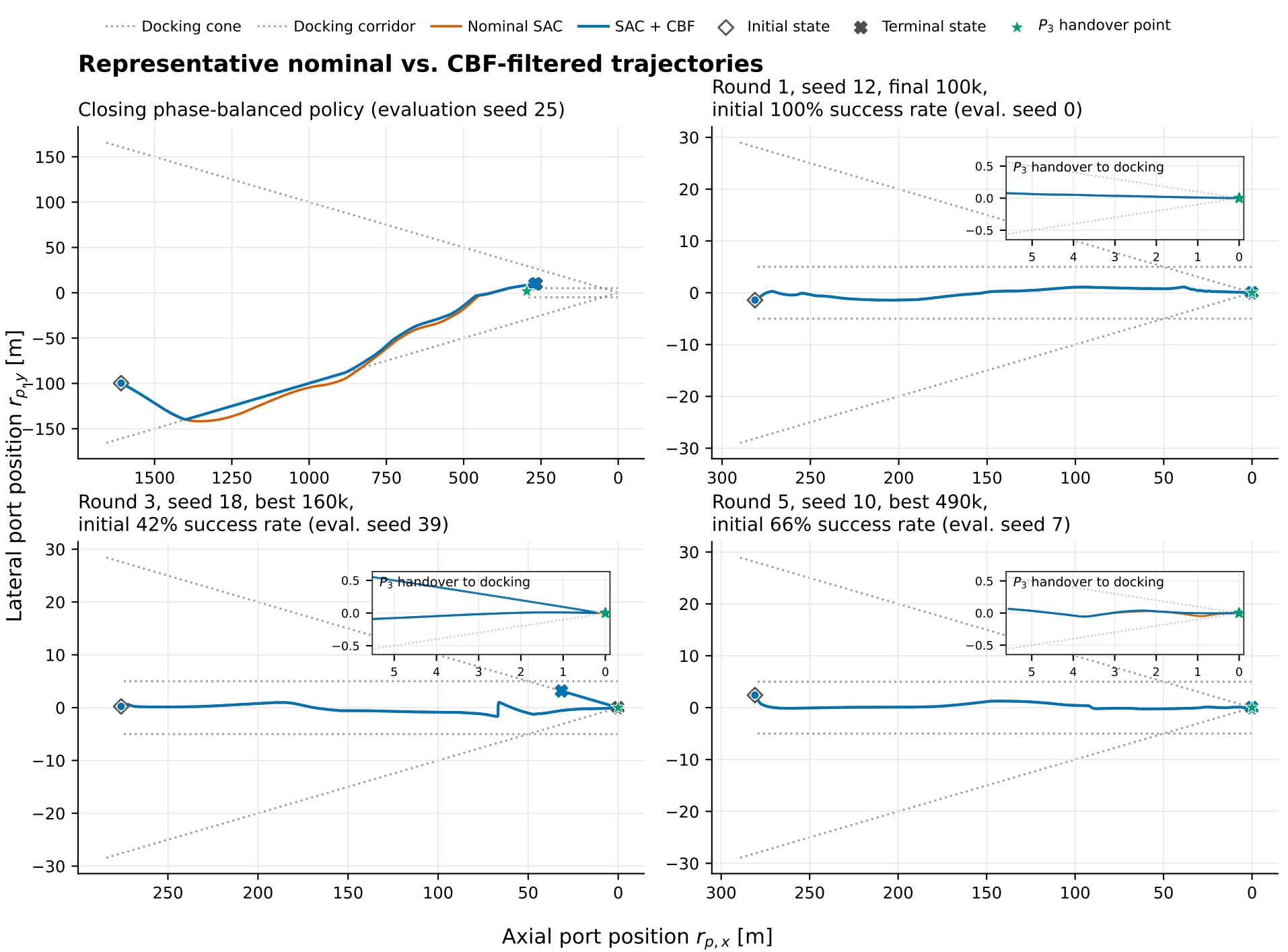
Docking cone
Docking corridor
Nominal SAC
SAC + CBF
Initial state
Terminal state
P_3 handover point
Representative nominal vs. CBF-filtered trajectories
Closing phase-balanced policy (evaluation seed 25)
Round 1, seed 12, final 100k,
initial 100% success rate (eval. seed 0)
Round 3, seed 18, best 160k,
initial 42% success rate (eval. seed 39)
Round 5, seed 10, best 490k,
initial 66% success rate (eval. seed 7)
P_3 handover to docking
Lateral port position $r_{p,y}$ [m]
Axial port position $r_{p,x}$ [m]